



Parthenocarpy In Cucurbits for Superior Seedless Fruit Production

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Abstract

Parthenocarpy, the development of fruits without fertilisation, enables the production of seedless, high-quality vegetables with consistent fruit set even under poor pollination or environmental stress. It occurs naturally through genetic and hormonal regulation or can be induced using plant growth regulators, distant hybridization, irradiated pollen, or polyploidy. Effective induction has been reported in crops such as cucumber, watermelon, muskmelon, pumpkin, bottle gourd, kakrol and pointed gourd using CPPU, GA₃, 2,4-D, and NAA. Molecular studies, including identification of QTLs such as Parth2.1, support marker-assisted breeding for enhancing parthenocarpic potential. Overall, parthenocarpy plays a major role in improving fruit quality, reliability of production, and suitability for protected cultivation in vegetable crops.

Introduction

Crops such as cucumber (*Cucumis sativus* L.), muskmelon (*Cucumis melo* L.), watermelon (*Citrullus lanatus* L.), squash and pumpkin (*Cucurbita* spp.), bitter melon (*Momordica charantia* L.), and bottle gourd (*Lagenaria siceraria* L.) are among the economically significant members of the Cucurbitaceae family. Like other monoecious species, cucurbits depend on suitable pollination for proper fruit development. Fruit yield can decline when pollinators are absent or when environmental conditions, such as insufficient light, excessive humidity, or high temperatures, are not ideal. Parthenocarpy offers a solution to overcome low fruit set under such challenging pollination conditions.

The demand for seedless fruits continues to grow because consumers prefer their improved flavour, extended shelf life, and ease of consumption. Parthenocarpic varieties also reduce labour costs by eliminating the need for artificial pollination, which is especially beneficial in cross-pollinated cucurbits.

Advantages of parthenocarpic vegetables

- Parthenocarpic vegetables produce seedless fruits that are more tender, uniform, and highly preferred in the market.
- They are capable of setting fruits even without pollination, ensuring reliable production throughout the season.
- These vegetables provide higher yields because fruit formation is not affected by the availability of pollinators.

- They maintain consistent fruit set under environmental stresses such as low temperature, high humidity, rain, or low light
- Parthenocarpy reduces the formation of misshapen or deformed fruits, resulting in better marketable quality
- They allow earlier, faster, and more uniform harvesting compared to normal varieties.
- The need for manual pollination or hormonal sprays is minimized, reducing labour and production costs.
- They are especially suitable for protected cultivation systems like polyhouses and greenhouses.
- Parthenocarpic fruits generally exhibit better firmness, improved quality, and longer shelf life during storage and transport.

Classification of Parthenocarpy:

Varoquaux *et al.* (2002) identified two forms of parthenocarpy:

- Natural/Genetic Parthenocarpy
- Induced/Artificial Parthenocarpy

Natural Parthenocarpy:

Natural parthenocarpy occurs when increased levels of endogenous hormones in the ovary trigger fruit development even without pollination or fertilization.

Genetic influences also contribute to parthenocarpy, as certain plant species possess inherited traits or specific genes that enable seedless fruit formation. These genes regulate hormone production, hormone-signalling pathways, and key steps in fruit development, thereby initiating fruit growth in the absence of fertilization.

Environmental conditions can further modulate parthenocarpic behaviour. Optimal factors such as favourable temperature, adequate light, moderate humidity, or mild stress may stimulate parthenocarpic fruit set, whereas adverse environmental conditions can suppress it.

Limited pollination is another cause of natural parthenocarpy. When pollinators like bees or other insects are absent or insufficient, some plants may initiate fruit development without fertilization to ensure reproductive continuity.

Hormonal imbalances within plant tissues may also induce parthenocarpy naturally. Elevated or altered levels of growth regulators such as auxins, gibberellins, and cytokinins play a central role in promoting ovary growth and fruit set without seed formation.

Artificial parthenocarpy:

Use of Plant growth regulators

Watermelon: Parthenocarpy in watermelon can be induced using CPPU (forchlorfenuron) at 0.5 ml/L. The application is carried out at anthesis to stimulate fruit set, resulting in completely seedless fruits suitable for premium fresh markets (Kawamura *et al.*, 2018).

Pumpkin: In pumpkin, parthenocarpic fruit development is achieved by applying Gibberellic acid (GA_3) at 150 ppm during anthesis. GA_3 promotes ovary enlargement without fertilization, producing up to 96.9% parthenocarpic fruits, indicating very high effectiveness (Sharif Hossain, 2015).

Muskmelon: Muskmelon responds strongly to cytokinin-based treatments. CPPU at 10 mg/L in combination with BA (Benzyladenine), significantly increases parthenocarpic fruit set by stimulating cell expansion and fruit growth even without pollination (Hayata *et al.*, 2000).

Bottle Gourd: Bottle gourd exhibits excellent response to CPPU application at 10–100 mg/L, either at anthesis or two days before anthesis. This treatment produces 100% seedless fruits, showing extremely high parthenocarpic induction (Jing, 1999).

Kakrol (Teasle gourd): Parthenocarpy in kakrol can be induced using 2,4-D at 50 ppm applied during anthesis, which effectively stimulates fruit development without fertilization. This method achieves 90% parthenocarpic fruit formation (Chowdhury *et al.*, 2007).

Pointed Gourd: Several plant growth regulators-2,4-D, NAA, CPPU, Fulmet, GA_3 , and TIBA can induce parthenocarpy in pointed gourd when applied at 25, 50, 100, or 200 ppm during anthesis. Responses vary with concentration: NAA at 200 ppm leads to abnormal seeds, while GA_3 at 200 ppm results in empty seeds, indicating partial parthenocarpy. (Hassan & Miyajima, 2019)

Distant hybridization

Spine Gourd: Parthenocarpy in spine gourd has been developed through distant hybridization using pollens from related species such as *Momordica charantia*, and *Luffa leucantha*. These interspecific pollinations enhanced natural parthenocarpic fruit development, achieving a maximum fruit set of 66%, indicating strong responsiveness. (Singh, 1978)

Pointed Gourd: In pointed gourd, distant hybridization using pollens from species such as *Momordica charantia* and *Luffa leucantha* successfully induced parthenocarpy. These crosses resulted in a high rate of seedless fruit formation, with the highest parthenocarpic fruit set recorded at 85%, which is considerably higher than that of spine gourd. (Singh, 1978)

Use of irradiated pollen

Soft X-irradiation has been used for inducing seedless fruit production in watermelon, with doses of 800–1000 Gy in the cultivar 'Fujihikari-TR' and 400–1000 Gy in 'Benikodama' resulting in fruits with slightly higher sugar content (Sugiyama & Morishita, 2000) and also using ray irradiation at 600 Gy and 800 Gy produced notable improvements, including a significant increase in total sugar content and enhanced carotenoid levels (Moussa & Salem, 2009). Another study on watermelon using soft X-irradiated pollen at 600 Gy demonstrated excellent results: the treated pollen, packed in vacuum or nitrogen and stored at either 4°C or 25°C for 14, 28, or 90 days, achieved 100% fruit set and successfully produced fully seedless fruits (Akutsu, 2018).

Polyploidy

Cucumber (Triploid and Autotetraploid Types): In cucumber (*Cucumis sativus*), parthenocarpy has been studied in relation to changes in ploidy levels. Triploid cucumbers ($2n = 3x = 21$) have been developed and reported to show stable parthenocarpic fruit formation, as noted by Chen *et al.* (2003). Another important cucurbit example is the autotetraploid cucumber line, developed in the cultivar Butchers Disease Resisting (BDR). This autotetraploid type ($2n = 4x = 28$) was produced using 0.2% colchicine treatment, as reported

by Grimby (1973). The colchicine treatment successfully induced chromosome doubling, leading to autotetraploidy, which in turn contributed to improved fruit development and supported parthenocarpic behaviour in cucumber.

Watermelon (Triploid Seedless Type): In watermelon (*Citrullus lanatus*), parthenocarpy and seedlessness are closely associated with triploid breeding. Triploid watermelon plants ($2n = 3x = 33$) are obtained by crossing autotetraploid lines with diploid parents. These triploid plants are functionally sterile and naturally produce seedless fruits due to failure of normal embryo development. According to Kihara (1951), the triploid condition enhances several desirable horticultural traits, including higher sugar content, increased number of fruits per plant, and thinner rind. These improvements make triploid seedless watermelons highly preferred for commercial cultivation and consumption.

Molecular markers and mapping of parthenocarpy

Parthenocarpy in cucumber (*Cucumis sativus*) has been well studied through QTL mapping, where a major-effect QTL named Parth2.1 along with six minor-effect QTLs was identified as controlling parthenocarpic fruit development. The mapping population used for this study was derived from a cross between EC1 and 8419 s-1, consisting of 145 $F_2: 3$ families. Marker screening revealed that 133 SSR markers out of 1335 tested and 9 In Del markers out of 173 tested were polymorphic and utilized for constructing the linkage map. In total, seven novel QTLs were detected across chromosomes 1, 2, 3, 5, and 7, with the major QTL Parth2.1 located on chromosome 2, flanked by the markers SSR00684 and SSR22083. This major locus plays an important role in enhancing parthenocarpy in cucumber through marker-assisted breeding (Wu *et al.*, 2016).

Example of Parthenocarpic Cultivars/ Varieties

crop	Institute	Varieties
Cucumber	IARI, New delhi	Pusa seedless cucumber 6
	GBPUA & T, Pantnagar	Pant parthenocarpic cucumber -1, Pant parthenocarpic cucumber -2, Pant parthenocarpic cucumber -3, Pant Khira-2 & Pant Khira-3
	KAU, Thrissur	KPCH-1
Watermelon	IIHR, Bengaluru	Arka Madhura
	KAU, Thrissur	Shonima(Red) and Swarna (Yellow)
	IARI, New Delhi	Pusa Bedana
Ivy gourd	CIAH, Bikaner	AHIG-1
Pointed gourd	IIVR, Varanasi	IIVRPG-105

Conclusion

Parthenocarpy represents a valuable biological and technological tool for enhancing seedless fruit production in vegetables, particularly under conditions where pollination is limited or environmental stresses hinder normal fruit set. Both natural and induced parthenocarpy contribute to improved fruit quality, uniformity, and market preference while reducing dependence on pollinators and labour-intensive pollination practices. Advances in plant growth regulators, distant hybridization, irradiated pollen techniques, polyploid breeding, and molecular mapping-such as the identification of major QTLs-have significantly strengthened the development and improvement of parthenocarpic traits in crops like cucumber, watermelon, muskmelon, pumpkin, and gourds. As consumer demand for seedless, high-quality produce continues to rise, integrating parthenocarpy into breeding and crop management strategies offers great potential for achieving higher productivity, stability, and sustainability in vegetable cultivation.

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